# Non-Line-of-Sight Optical Camera Communication: A Survey of Research Frontiers and Technical Details

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**Abstract**

Non-Line-of-Sight Optical Camera Communication (NLOS-OCC) extends conventional optical camera communication beyond direct visibility by exploiting diffuse reflections, partial occlusions, and ambient-light scattering to recover data from light-emitting diode (LED) signals that never reach the camera along a free-space path. The central technical contradiction is that LEDs are modulated in the MHz range while consumer cameras sample at 30–60 frames per second, so a single frame cannot resolve individual symbols directly. This survey organizes the rapidly expanding NLOS-OCC literature into a five-layer research framework comprising twelve research points (**RP1**–**RP12**): physical-layer channel modelling, modulation and coding, signal recovery, hardware and receiver front-ends, networking, and applications. For each layer we distil the key technical mechanisms—Barker-code pilots, differential dual-bit processing, rolling-shutter stripe extraction, U-Net bit segmentation, generative-adversarial background/stripe separation, event-camera asynchronous detection, reconfigurable intelligent surfaces (RIS), MIMO and NOMA schemes, and learning-based smartphone OCC architectures. We further survey four emerging directions (**RP13**–**RP16**): UAV-mounted OCC, underwater and cross-medium OCC, semantic optical communication toward AI-native 6G, and retroreflective optical integrated sensing and communication (RO-ISAC). We further examine the recently approved IEEE 802.15.7a-2024 amendment whose PHY VII defines seven OCC modes targeting 200 m range and 350 km/h mobility. We close with a prioritized roadmap that traces the dependency chain from channel characterization through signal recovery and networking to end-user applications, identifying near-, mid-, and long-term research priorities.

**Index Terms**—Optical camera communication (OCC), non-line-of-sight (NLOS), visible light communication (VLC), rolling shutter, deep learning, event camera, MIMO, RIS, IEEE 802.15.7a, vehicular communication, Internet of Things, UAV, underwater OCC, semantic communication, integrated sensing and communication (ISAC).

## 1 Introduction

Optical Camera Communication (OCC) is a branch of visible light communication (VLC) in which the receiver is an image sensor rather than a photodiode [12]. By spatially resolving light sources, a camera can discriminate among multiple transmitters and reject ambient interference in a way that single-element receivers cannot. The cost of this spatial selectivity is a severe temporal sampling limitation: a commodity camera captures 30–60 frames per second, while the LED driving signal may be modulated at MHz frequencies. The rolling-shutter effect, whereby successive rows of the sensor are exposed at slightly different times, partially alleviates this by spreading a time-varying signal across the image rows, producing characteristic light-and-dark *stripes*; but it cannot match the throughput of photodiode-based VLC.

The challenge becomes even more acute when the line-of-sight path between the LED and the camera is blocked. In Non-Line-of-Sight OCC (NLOS-OCC), the camera must recover data from diffuse reflections off walls, the ground, or other surfaces, or from partially occluded sources seen through glass or foliage. The received optical power is drastically reduced, the signal is smeared by the bidirectional reflectance distribution function (BRDF) of the reflecting surface, and the background illumination is heterogeneous. Yet NLOS operation is essential for practical deployment: in vehicular scenarios vehicles are frequently obscured by other traffic; in indoor scenarios furniture and occupants block direct paths; and in smart-city deployments street furniture creates persistent occlusions.

This survey organizes the NLOS-OCC literature into a five-layer research framework (Fig. 1) and twelve core research points (**RP1**–**RP12**), summarized in Table 1, plus four emerging directions (**RP13**–**RP16**) surveyed in Section 7. The framework mirrors the protocol stack: the physical layer (**RP1**–**RP3**) covers channel modelling, modulation, and coding; the signal-recovery layer (**RP4**–**RP6**) covers noise suppression, deep-learning extraction, and event-camera detection; the hardware layer (**RP7**–**RP8**) covers smartphone

Table 1: The NLOS-OCC research framework: sixteen research points across six layers.

| Layer | RP | Focus |
|---|---|---|
| Physical (**RP1**–**RP3**) | **RP1** | NLOS-OCC channel modelling (BRDF, ray tracing, geometry) |
| | **RP2** | Modulation schemes (OOK, PWM, UFSOOK, CSK, CSS, N-pulse, adaptive) |
| | **RP3** | Channel coding (Raptor/RaptorQ fountain codes, RS, FEC) |
| Signal Recovery (**RP4**–**RP6**) | **RP4** | Noise suppression and adaptive thresholding |
| | **RP5** | Deep-learning signal recovery (U-Net, CNN, GAN, autoencoder) |
| | **RP6** | Event-camera / neuromorphic asynchronous detection |
| Hardware (**RP7**–**RP8**) | **RP7** | Smartphone OCC (Pro mode, RAW domain, deep-learning demodulation) |
| | **RP8** | RIS / ORIS for NLOS coverage enhancement |
| Network (**RP9**–**RP10**) | **RP9** | MIMO schemes (WDM-MIMO, SDM, 2×N, MIMO-NOMA) |
| | **RP10** | Heterogeneous fusion (OCC/BLE, OCC/WiFi, 6G, IEEE 802.15.7a) |
| Applications (**RP11**–**RP12**) | **RP11** | V2V / V2I / ITS vehicular systems |
| | **RP12** | Indoor VLP, IoT, ISAC, smart-city deployment |
| Emerging (**RP13**–**RP16**) | **RP13** | UAV-mounted OCC (drone receiver, 360° FOV) |
| | **RP14** | Underwater and cross-medium (air-water) OCC |
| | **RP15** | Semantic optical communication, AI-native 6G integration |
| | **RP16** | Retroreflective optical ISAC (RO-ISAC) |

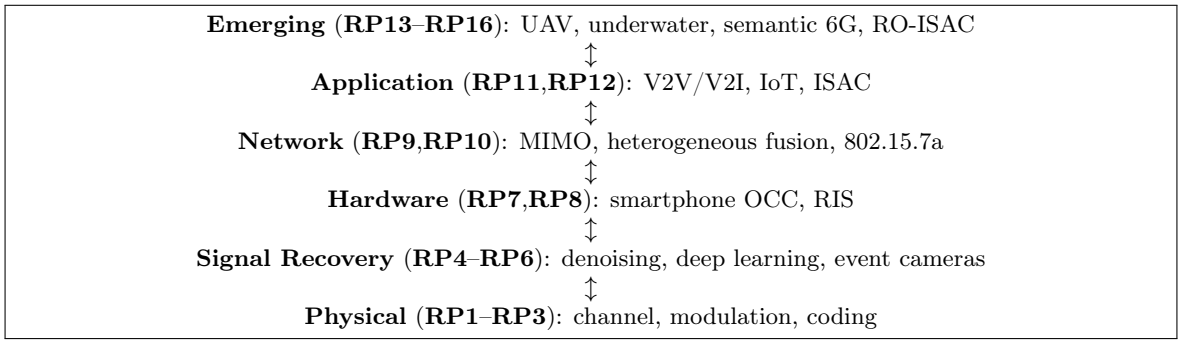



Figure 1: Five-layer NLOS-OCC research framework. Information flows upward; constraints propagate downward. The dependency chain **RP1** → **RP4/RP6** → **RP9** → **RP11** defines the critical research path.

receivers and RIS-enhanced links; the network layer (**RP9**–**RP10**) covers MIMO and heterogeneous fusion; the application layer (**RP11**–**RP12**) covers V2V/V2I/IoT systems; and the emerging layer (**RP13**–**RP16**) covers UAV, underwater, semantic, and ISAC frontiers.

The remainder of the paper follows the framework layer by layer (Sections 2–6), surveys four emerging research directions (Section 7), examines the IEEE 802.15.7a-2024 standard (Section 8), and closes with a prioritized roadmap and conclusions (Sections 9–10).

# 2 Physical Layer: Channel, Modulation, and Coding

## 2.1 NLOS-OCC Channel Modelling (RP1)

The NLOS-OCC channel departs fundamentally from the LOS VLC channel. Instead of a direct irradiance link, the received optical power is the integral of the LED radiance reflected by one or more surfaces whose BRDF may be Lambertian, specular, or mixed. Three modelling approaches dominate the literature.

**Analytical BRDF models.** A Lambertian reflector re-emits incident light uniformly in all directions with radiance proportional to the cosine of the exit angle. Real surfaces—asphalt, painted walls, foliage—deviate from this ideal, and the deviation directly affects the spatial distribution of received power across the image sensor. The pilot-aided NLOS-OCC work of Chen *et al.* [5] measured a 2.5 m NLOS link under heterogeneous background light, demonstrating that the pilot-based signal-logging and estimation (PSLE) scheme combined with enhanced visual compensation (EVC) can recover data where unaided thresholding fails entirely. The

experiment confirms that the reflected-signal contrast ratio is typically 5–15 dB below the LOS case, placing a hard floor on the achievable stripe signal-to-noise ratio.

**Ray-tracing simulation.** For vehicular MIMO channels, Liu and Dang [18] proposed a three-dimensional irregular-shaped geometry-based stochastic model (3D IS-GBSM) that captures the birth-death process of scattering clusters as vehicles move. The model parameters are derived from OpticStudio ray tracing and validated against measured power-delay profiles. The same ray-tracing approach underlies the MIMO V2V-VLC study of Yahia *et al.* [34], which simulates dual-transmitter headlight configurations and weather-dependent attenuation (clear, rain, fog). A key finding is that MIMO spatial multiplexing gain collapses rapidly as fog visibility drops below 100 m, motivating the adaptive and coding strategies discussed in Section 2.3.

**Pilot-based long-distance NLOS.** Dong *et al.* [6] achieved a 5 m NLOS-OCC link using a Barker-code pilot and a differential dual-bit processing (DDBP) algorithm. The Barker code's autocorrelation property lets the receiver locate the pilot in the stripe sequence even when the reflected signal is deeply buried in ambient light; DDBP then cancels the slowly varying background by subtracting consecutive bit windows. The 7% forward-error- correction (FEC) threshold was met at 5 m, extending the practical NLOS range beyond earlier pilot-free demonstrations.

## 2.2 Modulation Schemes (RP2)

OCC modulation must respect the camera's sampling constraints. The principal families are:

- **OOK / PWM.** On-off keying and pulse-width modulation are the simplest schemes. With rolling shutter, each symbol produces a band of bright or dark rows. The symbol rate is limited to roughly $f_s/2$ where $f_s$ is the effective row-sampling frequency (typically a few kHz).
- **UFSOOK.** Undersampled Frequency Shift OOK modulates at a frequency that aliases to a distinguishable low frequency under the camera's sampling, enabling per-frame demodulation without relying on rolling shutter. The MIMO extension [27] maps different frequencies to spatially separated LEDs, achieving a rudimentary spatial-multiplexing gain.
- **CSK / Color barcode.** Color Shift Keying encodes data in the colour coordinate of the emitted light. COBRA [10] demonstrated real-time colour-barcode streaming to smartphones, and the approach extends naturally to multi-colour LED arrays. CSK's chromaticity constraint (perceptual flicker avoidance) limits the practical constellation size.
- **Spatial modulation / PixNet.** PixNet [25] treats an LED array as a 2D spatial modulator, encoding blocks of bits in the on/off pattern of individual elements. This is the screen-camera analog of MIMO spatial modulation and achieves the highest raw rates among OCC schemes, but requires a well-resolved array image—difficult in NLOS conditions.
- **Adaptive modulation.** Shirvani *et al.* [29] investigated adaptive thresholding for V2V OCC where a mobile-phone camera receives LED taillight signals. The threshold is adjusted per-frame based on the measured stripe intensity statistics, accommodating the rapid variation in channel gain as the vehicles' relative position changes.

For NLOS links specifically, the modulation choice is constrained by the low contrast of the reflected signal. Pure OOK with a fixed threshold fails because the reflected intensity varies with surface geometry; adaptive and differential schemes (DDBP, PSLE) are essential.

## 2.3 Channel Coding (RP3)

Because the NLOS-OCC channel suffers deep fades from surface roughness, transient occlusions, and frame drops, rateless and fountain codes are particularly attractive: they generate as many encoded symbols as needed until the receiver acknowledges successful decoding, gracefully adapting to the unknown channel quality.

Raptor codes were applied to free-space optical communication by Lu *et al.* [19], and RaptorQ—a more robust variant with near-MDS (minimum distance separable) performance—was studied under atmospheric turbulence by Pernice *et al.* [26]. RaptorQ's advantage in the OCC context is its ability to recover the original block from any sufficiently large subset of received symbols, tolerating the bursty erasures that arise when a vehicle passes behind an obstruction and the camera drops several consecutive frames.

The IEEE 802.15.7a-2024 amendment (Section 8) adopts Reed-Solomon (RS) codes in its RS-OFDM mode, combined with convolutional or LDPC inner codes, providing a standardized alternative to the research-grade RaptorQ approach.

# 3 Signal Recovery Layer

## 3.1 Noise Suppression and Adaptive Thresholding (RP4)

In NLOS-OCC the signal of interest—the faint stripe pattern from a reflected LED—is superimposed on a spatially varying background comprising ambient illumination, other light sources, and the reflecting surface's own texture. Classical thresholding assumes a stationary background; in NLOS this assumption fails.

The pilot-aided approaches (**RP1**) already incorporate background cancellation (DDBP, EVC). Beyond these, adaptive per-stripe thresholding [29] estimates the local mean and variance from the stripe's own row profile and sets a threshold at the midpoint, yielding robust demodulation under vehicle motion. For quasi-NLOS screen-camera links through glass, Wang *et al.* [16] demonstrated that the glass reflection creates a secondary image that can be separated by disparity cues, recovering communication behind the glass. For long-distance outdoor links where atmospheric turbulence causes intensity fluctuations, Li *et al.* [15] proposed a Manchester-coding-based demodulation that exploits the guaranteed level transition in each symbol period, making the decoder robust to slow scintillation that would defeat a fixed threshold.

## 3.2 Deep-Learning Signal Recovery (RP5)

The last several years have seen a rapid migration from hand-crafted signal processing to learned pipelines. The principal architectures are:

**U-Net bit segmentation.** A U-Net semantic-segmentation network can be trained to produce a pixel-level binary mask classifying each pixel as "bright stripe" or "dark stripe" [24]. The encoder–decoder structure with skip connections preserves the fine spatial detail of the stripe edges, which a plain CNN classifier tends to blur. Once the mask is obtained, the bit sequence follows by simple row-major reading. This approach is particularly effective for NLOS reflections where the stripe boundaries are diffuse and the contrast is low.

**GAN-based background/stripe separation.** When the background is dynamic—moving vehicles, changing illumination, flickering displays—a fixed or slowly adaptive threshold cannot separate signal from background. A generative adversarial network can be trained to decompose the captured image into a stripe component and a background component, the discriminator enforcing that the generated background is indistinguishable from real background-only frames [17]. The separated stripe image is then demodulated with conventional or learned decoders. The GAN approach has been shown to reduce bit-error rate under motion and partial occlusion, achieving usable NLOS communication at 2.5 m with $10^{-4}$ BER in the remote-control demonstration of Ahmed *et al.* (DOI 10.1109/JPHOT.2023.3250455).

**Two-stage and end-to-end DNN.** A two-stage pipeline—LED detection followed by stripe classification—reduces the search space before demodulation, improving both accuracy and latency. End-to-end autoencoders that map bits to LED drive signals and images back to bits have been explored for LOS OCC and are a promising direction for NLOS, though training-data scarcity for the NLOS channel remains an obstacle. For multi-level Colour Shift Keying (CSK), Furukawa *et al.* [8] showed that a deep-learning equalizer can compensate the constellation distortion caused by hue shifts of the light source and the image sensor's internal colour processing, recovering the nominal CSK constellation points and improving demodulation accuracy—an important step toward high-spectral-efficiency OCC beyond binary modulation.

## 3.3 Event Cameras and Neuromorphic Detection (RP6)

The most architecturally radical approach to the sampling bottleneck is to abandon the frame-based camera entirely. An event camera (Dynamic Vision Sensor, DVS) fires an asynchronous pixel-level event whenever the log-intensity at that pixel changes by more than a threshold. Because the LED's high-frequency modulation produces continuous intensity changes, the event camera effectively samples the signal at ∼20 kHz—three orders of magnitude above frame-based cameras.

The neuromorphic NLOS-OWC work [21] (arXiv 2503.11226) is the first demonstration of event-camera-based NLOS optical wireless communication. Key results include an effective sampling rate of ∼20 kHz and a dynamic range of 120 dB, the latter being critical for NLOS where the reflected signal may span several orders

of magnitude as the reflecting surface geometry changes. The motion-aware event-camera OCC study [30] further shows that the asynchronous detection is inherently tolerant to transmitter and receiver motion, a property that is highly desirable for vehicular deployment.

The trade-off is that event cameras produce sparse, asynchronous data streams that require fundamentally different processing pipelines (spiking neural networks, event-to-frame conversion, or direct event-based decoding) compared to the image-centric pipelines of **RP4**–**RP5**.

# 4 Hardware Layer

## 4.1 Smartphone OCC: Pro Mode, RAW Domain, and Learning-Based Demodulation (RP7)

The ultimate deployment target for much OCC research is the commodity smartphone, whose camera is ubiquitous but whose image signal processor (ISP) is optimized for photography, not communication. The ISP applies auto-exposure, auto-white-balance, demosaicing, denoising, and JPEG/H.265 compression—each of which can destroy the faint stripe pattern that carries data.

**WinkLink / Xiao *et al.*** The MobiSys 2024 work of Xiao *et al.* [32] tackles the hardest practical case: optical camera communication when the LED emitter is *unseen* (outside the camera's field of view or occluded by complex, non-uniform backgrounds). The system employs a two-stage deep neural network. The first stage performs context-aware demodulation: it learns to separate the faint data-bearing stripe pattern from arbitrary background clutter (posters, textured walls, mixed illumination) that would defeat any fixed-threshold detector. The second stage reconstructs the transmitted bits from the cleaned signal. Crucially, the network is trained entirely on *synthesized* data —rendered stripes composited onto photographed backgrounds—yet generalizes to the real world: it was validated in 12 distinct environments and tolerates device motion during capture. Quantitatively, the system operates at $-20$ dB SNR, delivers a 5.8 dB SNR gain over the best baseline, and extends the effective communication distance by $5.5\times$ (reaching 11 m with a 10 W LED). This data-driven, learning-centric approach represents a paradigm shift from the hand-crafted signal-processing pipelines that dominate earlier OCC literature, and it directly addresses the NLOS "complex background" problem that is the central challenge of this survey.

**Passive and ambient-light approaches.** PassiveVLC [33] (MobiCom 2017) takes a complementary approach: rather than driving an LED, it modulates the reflectance of a passive tag illuminated by ambient light, enabling backscatter communication with zero transmit energy. LightSync [11] (MobiCom 2013) uses ambient light fluctuations for passive inter-device clock synchronization, a building block for any OCC system that must align transmitter and receiver timing without a shared reference.

**Pro mode and 4CASK-OCC.** Modern smartphone camera APIs (Android Camera2, iOS AVFoundation) expose manual exposure, focus, and ISO controls. Exploiting these "Pro mode" settings—short exposure, fixed white balance, high ISO—substantially improves stripe visibility. The 4CASK-OOC scheme (four-colour CSK) extends the colour-barcode idea to the smartphone's Bayer pattern, mapping symbols to quadrants of the CIE colour space to maximize perceptual-flicker-free throughput.

## 4.2 RIS / ORIS for NLOS Coverage (RP8)

When no reflecting surface is available, or when the natural reflection is too weak, a Reconfigurable Intelligent Surface (RIS) can be deployed to create an artificial NLOS path. An RIS is a planar array of sub-wavelength elements whose reflection phase can be controlled electronically, allowing the surface to steer an incident beam toward the receiver even when the transmitter and receiver are not mutually visible.

The RIS-ISAC study [2] formulates the joint beamforming design for an RIS-assisted integrated sensing and communication (ISAC) V2X system, minimizing the Cramér-Rao bound of the sensing estimate subject to SINR constraints on the communication links. Reported coverage enhancements are 15–20 dB in the NLoS region, transforming an unusable blocked link into one with margin to spare. For OCC specifically, an outdoor RIS (ORIS) can redirect a vehicle's headlight modulation toward a roadside camera that would otherwise be in shadow, bridging the coverage gap at intersections and in urban canyons.

# 5 Network Layer

## 5.1 MIMO Schemes (RP9)

Multiple-input multiple-output (MIMO) techniques increase throughput and reliability by exploiting the spatial dimension. In OCC, the camera's spatial resolution is a natural enabler: each LED in an array can carry an independent data stream, and the image sensor separates them by location.

**WDM-MIMO and SDM.** Wavelength-division multiplexing (WDM) assigns different colours (R/G/B) to different streams; space-division multiplexing (SDM) assigns different physical LEDs. Both are limited by the camera's colour separation and spatial resolution, respectively.

**2×N MIMO and ray tracing.** Yahia *et al.* [34] simulated a 2×2 MIMO V2V-VLC system using dual-transmitter headlight configurations and OpticStudio ray tracing, quantifying the channel cross-talk under clear, rain, and fog conditions. The results show that zero-forcing (ZF) spatial demultiplexing is adequate in clear weather but degrades as weather reduces the rank of the channel matrix.

**MIMO-NOMA.** The combination of MIMO with non-orthogonal multiple access (NOMA) is a powerful approach for V2V-VLC. In [20], a 2×2 MIMO system uses power-domain NOMA with successive interference cancellation (SIC), DCO-OFDM for bandwidth efficiency, and a novel normalized gain difference power allocation (NGDPA) scheme. The reported throughput improvement is 20% over OFDMA. A related beamforming study [36] applies NOMA with SIC and beamforming in the V2V context, showing further gains when the beamformer aligns the power-domain layers with the channel's spatial modes.

**MIMO survey.** Sejan *et al.* [28] provide a comprehensive survey of MIMO-VLC covering the full taxonomy (multiplexing, diversity, multi-user), machine-learning methods for channel estimation and demodulation, and future trends toward massive MIMO and intelligent reflecting surfaces.

## 5.2 Heterogeneous Fusion (RP10)

Pure OCC links are brittle: they fail in direct sunlight, in heavy fog, and when the camera's view is fully obstructed. Heterogeneous fusion combines OCC with a complementary radio technology to provide a robust overlay.

**OCC/VLC + RF.** Okasha *et al.* [23] propose a hybrid car-to-car MIMO VLC+RF system in which each vehicle alternates between a 100 m VLC hop and a 200 m RF hop, supporting up to 5 relay hops. The VLC hop achieves four times the throughput of the RF hop when the optical link is available, and the RF hop maintains connectivity when it is not.

**OCC/BLE and OCC/WiFi.** In indoor IoT scenarios, fusing OCC with Bluetooth Low Energy (BLE) or WiFi provides seamless handover: OCC delivers high-rate downlink data to a smartphone camera while BLE/WiFi handles the low-rate uplink and control channel. This split is natural because smartphones lack an uplink light source suitable for communication.

**6G and IEEE 802.15.7a.** The IEEE 802.15.7a-2024 amendment (Section 8) formalizes OCC as a PHY within the short-range optical wireless standard, providing the interoperability layer needed for heterogeneous 6G networks. The standard's relaying mechanism enables an OCC link to hand off to a neighbouring OCC or RF node, implementing the fusion architectures above in a standardized way.

# 6 Application Layer

## 6.1 Vehicular Communication: V2V, V2I, and ITS (RP11)

Vehicular visible light communication is the flagship application of OCC. Yamazato *et al.* [35] laid the foundation by demonstrating V2V and V2I links using vehicle LED headlights and taillights, highlighting the road-surface reflection channel that naturally arises in urban driving. Aly *et al.* [1] improved robustness with selection combining receiver diversity, selecting the best of multiple photodetector branches to combat the rapid fading of the vehicular channel.

For camera-based receivers, the V2V OCC system of Shirvani *et al.* [29] uses a mobile-phone camera pointed at the preceding vehicle's taillights, with adaptive thresholding to handle the changing channel. The MIMO-NOMA V2V-VLC system [20] pushes raw rates higher with 2×2 spatial multiplexing and power-domain superposition, while the hybrid VLC+RF architecture [23] extends range beyond a single hop. The RIS-ISAC V2X design [2] adds intelligent reflecting surfaces at intersections to maintain coverage when buildings block the direct path.

### 6.2 Indoor VLP, IoT, and ISAC (RP12)

Beyond vehicles, NLOS-OCC serves indoor visible light positioning (VLP), Internet of Things (IoT) sensor connectivity, and integrated sensing and communication (ISAC). Indoor VLP leverages the camera's angular resolution to estimate the position of an LED beacon from its image, achieving sub-metre accuracy even when the beacon is seen only as a diffuse reflection on a wall. IoT applications use low-rate OCC to read sensor values from LED indicators on appliances, with the smartphone camera as the universal reader. ISAC combines the communication function with a sensing function (presence detection, gesture recognition) using the same LED illumination, an approach that aligns with the RIS-ISAC framework of [2] and the AI-PHY provisions of the IEEE 802.15.7a standard.

## 7 Emerging Research Directions

Beyond the twelve research points of the core framework, four emerging directions are reshaping the NLOS-OCC landscape. These directions extend the framework upward (toward 6G semantic integration), outward (to UAV and underwater environments), and inward (to joint sensing-communication waveforms). We label them **RP13**–**RP16** to indicate their position as frontier extensions of the five-layer stack.

### 7.1 UAV-Mounted OCC (RP13)

Unmanned aerial vehicles (UAVs) equipped with cameras or LED payloads open a new deployment axis for OCC. Chang *et al.* [4] demonstrated the first wide-field-of-view (FOV) bidirectional UAV-assisted OWC system using the under-sampled phase shift on-off keying (UPS-OOK) format and a light-diffusing-fibre transmitter that radiates over 360°, enabling a hovering UAV to serve ground receivers at any azimuth. The long-short-term-memory neural network (LSTM-NN) decodes the rolling-shutter pattern from arbitrary viewing angles, removing the pointing constraint that burdens conventional OCC.

For agricultural IoT, Kondo *et al.* [14] proposed a drone-based OCC sensor network in which LED panels on ground sensor nodes transmit soil-moisture and temperature data to a drone-mounted camera. A trajectory-control algorithm reduces total flight time by 30 % versus a shortest-path baseline, and YOLOv5-based LED detection with simple OOK modulation achieves BER $< 10^{-3}$ in a real-world leaf-mustard farm trial. This paradigm—low-cost, radio-silent, infrastructure-free—is attractive for underdeveloped and radio-quiet zones.

### 7.2 Underwater and Cross-Medium OCC (RP14)

Underwater OCC (UWOCC) inherits the NLOS challenge in an extreme form: the water column introduces wavelength-dependent absorption, particulate scattering, and turbulence that jointly attenuate and smear the optical signal. Hamidnejad and Gholami [9] developed a comprehensive underwater MIMO OCC channel model that captures these effects and showed that spatial modulation—exploiting the camera's ability to resolve multiple LED transmitters—retains its throughput advantage even in turbid water.

At the air-water interface, Tsai *et al.* [31] combined the UAV and underwater themes: a UAV-mounted camera receives OCC signals transmitted from an underwater source, crossing the wavy, cloudy water surface. An LSTM-NN mitigates the surface-fluctuation-induced link outage, demonstrating that cross-medium NLOS OCC—from a submerged sensor to an aerial camera—is feasible. This opens paths for autonomous-underwater-vehicle (AUV) telemetry retrieval without recovering the vehicle to the surface.

### 7.3 Semantic Optical Communication and AI-Native 6G (RP15)

Semantic communication shifts the fidelity criterion from bit-level reconstruction to task-relevant meaning, trading redundant detail for robustness. For underwater image transmission, Nennouche *et al.* [22] proposed a variational-autoencoder (VAE) semantic architecture that jointly optimizes source coding and channel adaptation; across varying channel conditions the semantic approach achieves superior image reconstruction over longer ranges than traditional compression plus error-correction coding. The principle transfers directly

to NLOS-OCC: a semantic encoder can prioritize safety-critical vehicular messages or structural-inspection features, transmitting only the meaning-bearing content that the degraded NLOS channel can support.

At the architectural level, Bostan [3] surveyed cross-layer AI-native 6G designs that integrate semantic communication, ISAC, RIS, and optical wireless communication (OWC) as complementary programmable physical-layer technologies. The five-domain reference architecture—physical, control/user-plane, edge-intelligence, AI-native orchestration, and semantic/task—positions OWC (including OCC) as a short-range, high-capacity, interference-free physical layer that semantic encoders can exploit when RF bands are congested. For NLOS-OCC this means the camera receiver may eventually feed a semantic decoder that extracts task-level information directly from the striped image, bypassing the bit-level pipeline entirely.

### 7.4 Retroreflective Optical ISAC (RP16)

Integrated sensing and communication (ISAC) over the optical band is a rising 6G theme. Retroreflective optical ISAC (RO-ISAC) pairs an interrogating transceiver with a passive corner-cube retroreflector (CCR) that returns light along the incident path, enhancing the reflected signal strength that is the Achilles' heel of NLOS links. Du *et al.* [7] designed a flexible waveform embedding an OFDM data signal within a maximum-length-sequence (MLS) sensing frame, jointly delivering communication and ranging/sensing on a single optical carrier. The CCR's strong back-reflection relaxes the NLOS power budget, and the OFDM-MLS embedding avoids the separate-sensor overhead of conventional ISAC. For NLOS-OCC this is significant: a retroreflector placed on a vehicle or infrastructure element can return enough light for a camera to decode both the data and a sensing measurement (range, velocity) from a single captured frame, merging **RP1** channel characterization with **RP12** ISAC in one physical mechanism.

## 8 The IEEE 802.15.7a-2024 Standard

The IEEE 802.15.7a-2024 amendment [13], approved 2024-12-11 and published 2025-02-14, defines PHY VII—a dedicated OCC physical layer with seven modes designed to span the full range of applications from low-rate IoT to high-rate vehicular communication:

1. **RS-OFDM** — Reed-Solomon outer code with OFDM, up to 9.6 Mbps, the highest-rate mode.
2. **MIMO-COOK** — MIMO with colour-orthogonal OOK, using the R/G/B channels as independent streams.
3. **O-NOMA** — Optical NOMA with power-domain superposition and SIC, supporting multi-user access on a single optical channel.
4. **MIMO-OOK** — Spatial-multiplexing MIMO with per-LED OOK, the camera resolving the LEDs by position.
5. **HOOK-OFDM** — Hybrid OOK with OFDM, combining the robustness of OOK with the spectral efficiency of OFDM.
6. **HS2PSK-OFDM** — Hybrid S2-PSK (spatial two-phase shift keying) with OFDM, a spatial-colour hybrid.
7. **BPPM** — Binary pulse position modulation, the most robust mode for low-SNR NLOS channels.

The standard targets a 200 m communication range and 350 km/h mobility, encompassing highway V2V scenarios. Two forward-looking provisions are notable: an *AI-PHY* framework that allows machine-learning-based transceiver components (codifying the deep-learning approaches of **RP5** within the standard), and a *relaying mechanism* that enables multi-hop OCC links—the standardized counterpart of the hybrid VLC+RF fusion in [23]. The base standard [12] defines PHY I–III (VLC with photodiode receivers) and PHY IV–VI (OCC), against which the amendment is defined.

## 9 Open Challenges and a Prioritized Roadmap

The literature reveals a clear dependency chain: **RP1** → **RP4**/**RP6** → **RP9** → **RP11**. Accurate channel models (**RP1**) define the signal characteristics that recovery algorithms (**RP4**, **RP6**) must handle; recovery performance determines whether MIMO and network-level techniques (**RP9**) can operate on the recovered symbols; and the network layer enables the applications (**RP11**) that justify the effort. We prioritize the research points accordingly.

Table 2: Prioritized research roadmap.

| Horizon | RPs | Rationale |
|---|---|---|
| Near-term | **RP4**, **RP7**, **RP12** | Denoising + smartphone = deployable demos |
| Mid-term | **RP1**, **RP5**, **RP9**, **RP11** | Channel + DL + MIMO ⇒ V2X |
| Long-term | **RP6**, **RP8**, **RP10** | Event cameras, RIS, 6G fusion |
| Frontier | **RP13**, **RP14**, **RP15**, **RP16** | UAV, underwater, semantic 6G, RO-ISAC |

**Near-term (RP4, RP7, RP12).** The fastest path to a deployable NLOS-OCC demonstration combines adaptive thresholding (**RP4**) with a smartphone receiver (**RP7**) in an indoor IoT or VLP scenario (**RP12**). The required components exist; the work is integration, robustness testing, and API-level access to the RAW sensor stream (as demonstrated by Xiao *et al.* [32], whose two-stage DNN achieves robust demodulation behind complex, unseen backgrounds).

**Mid-term (RP1, RP5, RP9, RP11).** The vehicular use case (**RP11**) demands accurate NLOS channel models (**RP1**), deep-learning recovery robust to motion and weather (**RP5**), and MIMO multiplexing for adequate throughput (**RP9**). The IEEE 802.15.7a modes provide the interoperability layer; the research challenge is closing the gap between laboratory demonstrations and the 200 m / 350 km/h targets of the standard.

**Long-term (RP6, RP8, RP10).** Event cameras (**RP6**) dissolve the sampling bottleneck but require a fundamentally new processing stack; RIS (**RP8**) adds coverage but introduces control-plane complexity; and heterogeneous 6G fusion (**RP10**) requires cross-layer orchestration of OCC, RF, and sensing. These are the directions with the highest potential payoff and the highest uncertainty.

**Frontier (RP13, RP14, RP15, RP16).** Four emerging directions extend NLOS-OCC beyond its terrestrial, bit-level origins. UAV-mounted OCC (**RP13**) [4, 14] adds a mobile aerial receiver that can serve infrastructure-free sensor networks; the challenge is trajectory optimization and wide-FOV optics. Underwater and cross-medium OCC (**RP14**) [9, 31] must contend with wavelength-dependent absorption and the turbulent air-water interface, but enables AUV telemetry and ocean-IoT links. Semantic optical communication (**RP15**) [3, 22] reframes the receiver as a meaning extractor rather than a bit decoder, trading fidelity for task-level robustness—a natural fit for the bandwidth-limited NLOS channel. Retroreflective optical ISAC (**RP16**) [7] merges communication and sensing on a single optical carrier using passive retroreflectors, directly addressing the NLOS power-budget deficit. These directions are at the concept-to-prototype stage; their convergence with the core framework will define the next generation of NLOS-OCC research.

**Cross-cutting challenges.** Three issues cut across all horizons: (i) *training data* for deep-learning NLOS-OCC is scarce and expensive to collect, motivating synthetic data generation via ray tracing and GAN-based augmentation; (ii) *synchronization* between the LED transmitter and the camera receiver remains non-trivial, especially under motion, and LightSync [11]-style ambient-light synchronization deserves renewed attention; and (iii) *standardization compliance* —ensuring that research prototypes can interoperate under the 802.15.7a-2024 framework—will determine whether the field's advances translate into deployed systems.

# 10 Conclusion

Non-Line-of-Sight Optical Camera Communication sits at the intersection of optical wireless, computational imaging, and mobile computing. Its core challenge—recovering MHz-modulated data from a reflection-challenged, ambient-light-flooded, 30-fps image stream—has been attacked from complementary directions: pilot-aided channel modelling and coding at the physical layer; U-Net and GAN-based deep-learning recovery at the signal layer; learning-based smartphone demodulation at the hardware layer; MIMO-NOMA and hybrid RF fusion at the network layer; and event cameras that rethink the sensing primitive itself. The IEEE 802.15.7a-2024 standard now provides the interoperability framework, and the dependency chain **RP1** → **RP4/RP6** → **RP9** → **RP11** identifies where the next advances must come from. Four emerging directions—UAV-mounted OCC (**RP13**), underwater and cross-medium OCC (**RP14**), semantic optical communication toward AI-native 6G (**RP15**), and retroreflective optical ISAC (**RP16**)—extend the framework to aerial, aquatic, semantic, and joint-sensing frontiers. The near-term opportunity is the smartphone-based indoor

demo; the mid-term prize is the vehicular V2X link at the standard's 200 m / 350 km/h target; and the long-term frontier is the event-camera-equipped, RIS-assisted, 6G-fused, semantically-aware NLOS-OCC network spanning air, water, and ground.

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